LAB ASSIGNMENT 6

Bias-Variance Tradeoff and Model Comparison

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Q1: Bias-Variance Tradeoff in Polynomial Regression

A synthetic regression dataset of 700 samples was generated with

$$x \sim U(-2, 2)$$

and target function

$$y = \sin(\pi x) + 0.3x^2 + \epsilon,$$

where $\epsilon \sim \mathcal{N}(0, 0.5)$.

To clearly demonstrate the bias-variance tradeoff, a small training subset of 80 samples was used, while the remaining samples formed the test set. Polynomial regression models of varying degrees up to 25 were trained using the closed-form normal equation with feature standardization.

Observations

- For low polynomial degrees, both training and testing mean squared error (MSE) were high, indicating underfitting and high bias.
- As model degree increased, testing error decreased and reached a minimum at intermediate complexity.
- For higher degrees, training error continued to decrease while testing error increased, demonstrating overfitting and high variance.

Conclusion

The experiment clearly illustrates the bias-variance tradeoff:

- Increasing model complexity reduces bias.
- Excessive complexity increases variance and harms generalization.

Q2: Comparison of L1 and L2 Regularization in Logistic Regression

A synthetic binary classification dataset with 600 samples and 5 features was generated. The decision function included both linear and nonlinear components. Logistic regression was trained using gradient descent for 100 epochs under three settings:

1. No regularization
2. L2 regularization
3. L1 regularization with soft-thresholding

Without Regularization

The model achieved high training and testing accuracy with a small generalization gap, indicating good separability of the dataset.

L2 Regularization

As the regularization parameter λ increased:

- Testing accuracy gradually decreased.
- Weight magnitudes were smoothly shrunk toward zero.
- Large λ values caused underfitting.

L2 regularization controlled model complexity without inducing sparsity.

L1 Regularization

With increasing λ:

- The number of non-zero coefficients decreased.
- For sufficiently large λ, all feature weights became zero.
- The classifier reduced to a constant predictor with approximately 50 percent accuracy.

L1 regularization promoted sparsity and effectively performed feature selection.

Conclusion

- L2 regularization reduces variance through weight shrinkage.
- L1 regularization induces sparsity and performs implicit feature selection.
- Excessive regularization leads to underfitting.

Q3: Effect of Class Priors in Gaussian Naive Bayes Classification

A two-dimensional dataset was generated with 400 samples per class using Gaussian distributions with identical covariance matrices:

$$\Sigma = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1.5 \end{bmatrix}.$$

A Gaussian Naive Bayes classifier was implemented. Although the data was generated with correlated features, the classifier assumed conditional independence between features and used class-specific variances.

Two prior settings were examined:

1. Equal priors: $P(C_0) = 0.5$, $P(C_1) = 0.5$
2. Unequal priors: $P(C_0) = 0.8$, $P(C_1) = 0.2$

Equal Priors

The decision boundary was approximately balanced between the two clusters. Both training and testing accuracy were high. The confusion matrix showed low false positive and false negative rates.

Unequal Priors

When the prior probability of Class 0 was increased:

- The decision boundary shifted toward the Class 1 region.
- False positives decreased.
- False negatives increased.
- The classification of the test point changed due to prior adjustment.

Conclusion

Class priors directly influence posterior probabilities and shift the decision boundary. In real-world applications such as fraud detection and medical diagnosis, incorporating correct prior probabilities is essential when dealing with class imbalance.

Overall Summary

- Q1 experimentally demonstrated the bias-variance tradeoff.
- Q2 highlighted the distinct effects of L1 and L2 regularization.
- Q3 showed how prior probabilities alter classification boundaries and error distribution.